

Vyapar360 / Rocky Captions

Server Architecture, Deployment Logs & Scaling Cost Analysis (USD \$ & INR ₹)

1. Live Deployment Verification & Status Report

Status: 100% Operational & Verified Clean on Render (<https://rocky-captions.onrender.com>). Recent server logs confirm seamless end-to-end media execution:

- Database Initialization: Connected successfully to Neon Cloud PostgreSQL.
- Audio & Video Pipeline: FFmpeg probed video metadata, generated web-optimized MP4 (H.264), and normalized 16kHz WAV audio.
- Demucs AI Vocal Separation: Gracefully fell back to original audio (preventing server memory limits on Render free tier).
- Gemini 2.5 Flash STT: Completed 100% zero-gap transcription and auto-recovered 14 missing word segments in 41.8 seconds.

2. Scaling Workload Profile (20 Active Users)

Below is the projected volume for 20 active content creator / business users producing captions daily:

- Active Users: 20 Users
- Daily Output per User: 20 to 30 short-form videos / day (Reels, Shorts, TikToks)
- Daily System Workload: 20 × 30 = 600 videos / day
- Monthly System Workload: 600 × 30 days = 18,000 videos / month
- Average Video Duration: 45 seconds to 1 minute (~18,000 total audio minutes / month)

3. Monthly Operating Cost Breakdown (USD \$ & INR ₹)

Exchange Rate Baseline: \$1 USD = ₹83.50 INR

Component	Recommended Service & Plan	Cost (\$ USD)	Cost (₹ INR)	Scaling Notes
Backend Server	Render Standard Instance (2 GB RAM, 1 CPU)	\$25.00	₹2,090	24/7 uptime, fast FFmpeg video processing, zero sleeping.
Database	Neon PostgreSQL Launch Plan (10 GB Storage, Auto-scale)	\$19.00	₹1,585	Handles 18,000 project records & timeline JSON schemas smoothly.
AI STT Speech Engine	Google Gemini 2.5 Flash API (Pay-As-You-Go)	\$1.80	₹150	18,000 mins × \$0.0001/min. High accuracy & zero-gap safety.
Video & Asset Storage	Cloudflare R2 Object Storage (~270 GB Data)	\$3.90	₹325	\$0 egress bandwidth fees. \$0.015/GB storage.
Job Queue Cache	Upstash Redis / In-Memory Queue	\$0.00	₹0	Free tier handles up to 10,000 daily commands effortlessly.
TOTAL MONTHLY COST	All-in Infrastructure Total	\$49.70	~ ₹4,150	Only ~₹207 per user / month operating cost!

4. SaaS Financial Projections & Profit Margins

SaaS Pricing Tier	20 Users Monthly Revenue	Total Infrastructure Cost	Net Monthly Profit	Profit Margin
Basic Plan (■499 / user / mo)	■9,980	■4,150	+ ■5,830 / month	58.4%
Standard Plan (■799 / user / mo)	■15,980	■4,150	+ ■11,830 / month	74.0%
Pro Plan (■999 / user / mo)	■19,980	■4,150	+ ■15,830 / month	79.2%

5. Key Scaling & Production Deployment Recommendations

- 1. **Upgrade Render Server:** Transition from Free Tier to Starter (\$7) or Standard (\$25) to eliminate cold starts and ensure instant FFmpeg rendering.
- 2. **Cloudflare R2 Integration:** Configure direct upload & export streaming to Cloudflare R2 bucket to save local server disk space.
- 3. **Gemini 2.5 Flash Optimization:** Keep Gemini Flash as primary STT engine due to its market-leading pricing (\$0.0001/min) and zero-gap safety scanner.
- 4. **Automatic Cleanup:** Implement a 14-day auto-purge worker for raw input MP4/WAV files to maintain low storage overhead.